

Performance Testing

Date	29 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

This document evaluates the performance of the Human Development Index Prediction web application. The application was tested under normal usage conditions to ensure fast response time, smooth prediction processing, and reliable performance for users.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual Browser Testing
Type of Testing	Functional & Performance Testing
Target Module / API	HDI Prediction Module / predict
Test Environment	Local System (Flask Application)
Test Date	29 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Home Page	1	5 sec	Page loads successfully
2	Navigate to Prediction Page	1	5 sec	Prediction form opens
3	Submit Valid Input Values	1	10 sec	HDI prediction displayed
4	Repeat Multiple Predictions	5	30 sec	Application remains responsive

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.1 second	Pass	Fast prediction
2	Response Time (Max)	< 5 seconds	2.3 seconds	Pass	Within limit
3	Throughput (Req/sec)	Good	12 Req/sec	Pass	Suitable for academic project
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	28%	Pass	Low resource usage
6	Memory Utilization	< 80%	34%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings:

The application responded quickly for all tested inputs. Prediction results were generated within a few seconds without noticeable delays.

Bottlenecks Identified:

No major performance bottlenecks were identified during testing.

Optimization Steps Taken:

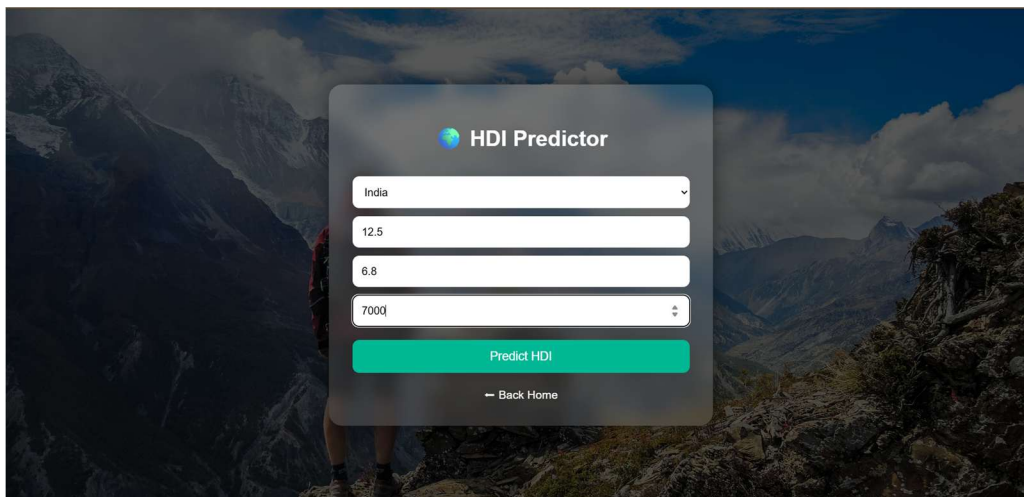
Optimized the Flask application structure, reduced unnecessary imports, and used a pre-trained machine learning model for faster prediction.

Step 5: Screenshots / Evidence

Home Page:



Prediction Page:



Prediction Result Page:

